School of Computer Science

GRADUATION PROJECT DOCUMENTATION

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Slim as Fit

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Abstract

The wearable fitness technology industry has experienced significant growth in recent years; however, here's still a clear gap when it comes to strength training. Existing products which primarily target cardio-based activities often lack accuracy in monitoring weightlifting performance. To address that gap, the Slim as Fit application is proposed as a comprehensive solution designed to advance strength training methodologies. Slim as Fit incorporates an IMU sensor within a small, human-friendly, wearable design that prioritizes comfort and ease of use. Unlike other fitness tracking devices, it employs a magnetic direct-attach mechanism for integration with weightlifting equipment such as dumbbells and barbells and even machine weights. This approach enables precise automated counting and differentiation between concentric and eccentric movements. The hardware interfaces with a mobile application via Bluetooth, which includes a gamified biofeedback system that delivers audio-visual feedback on lifting speed and consistency. By combining accurate hardware with advanced software tools, the proposed application addresses the lack of supervision in autonomous lifting environments, thereby enhancing operational safety and providing reliable analytics for lifting performance.

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List of Acronyms / Abbreviations

Abbreviations	Word
1RM	**O**ne **R**epetition **M**aximum
AI	**A**rtificial **I**ntelligence
BLE	**B**luetooth **L**ow **E**nergy
ERD	**E**ntity **R**elationship **D**iagram
FR	**F**unctional **R**equirement
IMU	**I**nertial **M**easurement **U**nit
MCU	**M**icrocontroller **U**nit
NFR	**N**on-**F**unctional **R**equirement
PCB	**P**rinted **C**ircuit **B**oard
Reps	**R**epetitions
RPE	**R**ate of **P**erceived **E**xertion
SQL	**S**tructured **Q**uery **L**anguage
TUT	**T**ime **U**nder **T**ension
UML	**U**nified **M**odeling **L**anguage
SQG	**S**urvey **Q**uestion **G**raph

Chapter 1: Introduction

1.1 Project Background & Overview

Over the last ten years, wearable fitness devices and health apps have become commonplace in daily wellness Routens're most of these tool's center on aerobic exercises running, cycling, walking while strength training and resistance work receive far less attention. This creates a noticeable void in available monitoring technology [1][2].

Slim as Fit was developed to fill this void. Rather than using wrist-worn sensors that track arm movement indirectly, our system places a small sensor unit directly onto workout equipment like barbells, dumbbells, or weight machines [3]. This sensor contains an Inertial Measurement Unit (IMU) that captures the actual motion of the equipment itself. The direct attachment allows for accurate tracking of how many reps you complete, how fast you move through each phase [5][6], your range of motion, and whether your form stays consistent throughout the set [7].

The system has two main parts. First, there's a physical sensor that mounts to your equipment and watches your movements in real time. Second, there's a mobile app that receives this data wirelessly through Bluetooth. But the app does more than just display numbers it manages your entire workout. Guides you through proper tempo using rhythm cues, shows you how to perform exercises correctly [8], gives you immediate feedback on your performance, and tracks your progress over weeks and months [9].

What makes Slim as Fit effective is how it brings together accurate physical sensing, smart data analysis, and user-friendly design [10]. Think of it as having a spotter who watches every rep and helps you train more systematically. It works for competitive athletes who want detailed metrics, beginners who need guidance, and older adults who prioritize safety. While fitness trackers have become increasingly popular, they still struggle to meet the specific demands that come with lifting weights.

1.2 Problem Statement

Despite the advancements of wearable fitness technology, a significant 'data gap' exists in the field of strength training and resistance exercise. Current market solutions, primarily wrist-based wearables and manual logging apps, fail to provide the precision and real-time guidance necessary for effective muscle hypertrophy (growth). A very critical flaw in these wrist-centric designs is their inability to track lower-body exercises where the upper body remains static. During movements like the Leg Press or Seated Squat, the user's wrists are often rested on the body; consequently, standard wearables fail to detect motion, registering high intensity sets as periods of inactivity. The core problems addressed are the following.

1.2.1. Lack of Strength Metrics in Wearables

Current smartwatches and fitness bands are primarily designed for cardio-based activities (e.g., walking, running, cycling) and fail to capture the actual kinematics of resistance training.

1.2.2. Lack of Tempo Control and Uncontrolled Lifting Behavior.

Proper muscle growth (hypertrophy) relies heavily on Time Under Tension (TUT). Many gym-goers, particularly beginners and "ego lifters," perform Reps too quickly or with jerky motions. Existing technologies track that a movement happened, but they do not monitor how it happened, failing to provide feedback on the rhythm or speed of the lift.

1.2.3. Neglect of the Eccentric Phase

In strength training, the lowering phase (eccentric) is just as important for hypertrophy as the lifting phase (concentric). Most trackers only count the "hit" of a rep. This ignores the common mistake of "dropping" the weight without control, which increases injury risk and reduces the metabolic stress required for progress.

1.2.4. Absence of Real-Time Form Correction

Beginners and seniors often lack the confidence or knowledge to perform exercises safely. Without a personal trainer, these users are at risk of injury. Current apps offer

static video tutorials, but they cannot provide active feedback or virtual coaching to alert a user when their form becomes shaky, or their speed drops due to fatigue.

1.2.5. Low Engagement and Retention

For many users, data logging is a tedious chore. There is a lack of interactive gamification in strength training that converts raw, technical movements into an engaging experience, leading many users to abandon their fitness routines due to a lack of immediate motivation.

1.2.6. Survey of User Problems

To better understand the challenges faced by gym users during resistance training, a survey was conducted with participants of varying fitness levels. The survey aimed to identify common difficulties related to workout tracking, repetition accuracy, and performance monitoring.

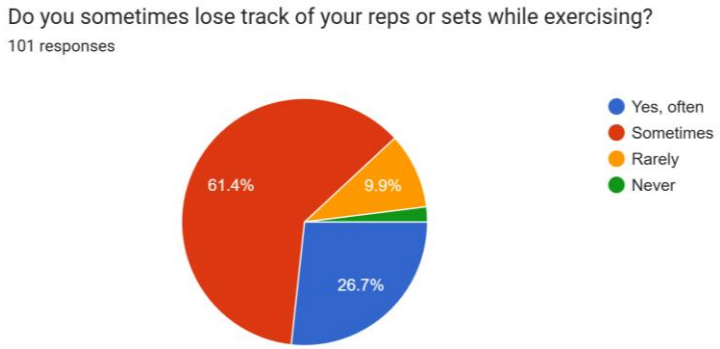

Figure 1.1 Losing track of counting problem (SQG1)

As shown in Figure 1.1, the majority of participants reported losing track of their Reps during workouts, with 61.4% indicating that this occurs sometimes and 26.7% reporting frequent difficulty.

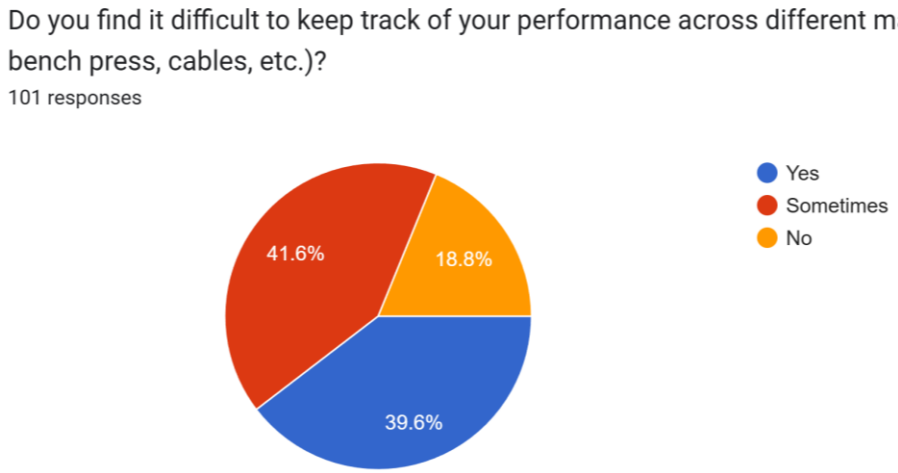

Figure 1.2 Disorientation and feeling lost (SQG2)

Figure 1.2 shows that 60.4% of respondents experience difficulty tracking performance across different gym machines, indicating a lack of unified workout tracking solutions.

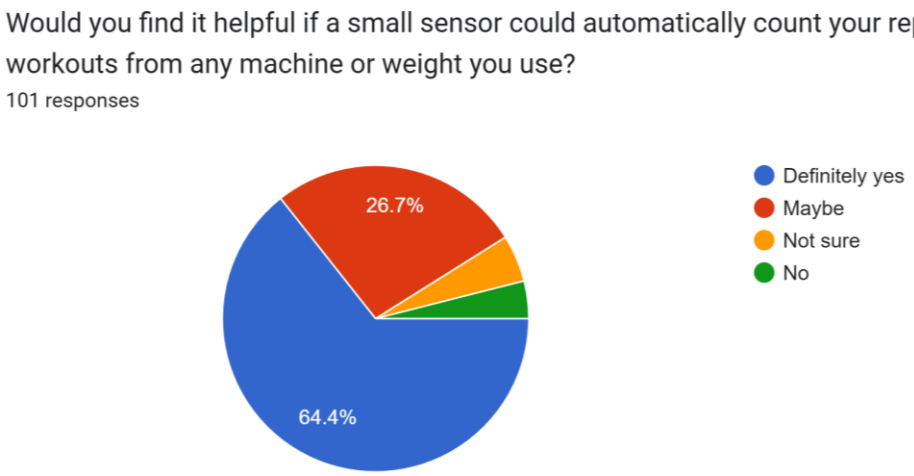

Figure 1.3 Product Demand Analysis (SQG3)

According to Figure 1.3, 64.4% of participants expressed definite interest in an automated rep-counting device, while only a small percentage indicated no interest. This supports the feasibility and relevance of the proposed system.

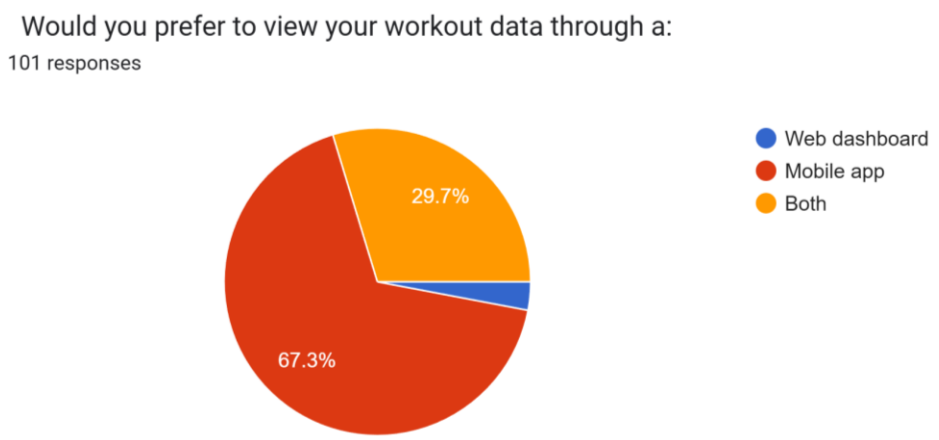

Figure 1.4 preference of device (SQG4)

The results conducted in Figure 1.4 indicate a strong preference for mobile applications (67.3%), validating the decision to design the system as a mobile-first application.

Based on the analysis of the survey responses, several important issues were identified among gym users, including difficulty in tracking Reps and sets, inconsistency in monitor the user performance across different machines, and the lack of real-time feedback during resistance training. Additionally, many users expressed strong interest in an automated and unified workout tracking solution, particularly through a mobile application.

1.3 Project Objectives

The challenges previously mentioned highlighted the need for a smart and integrated system capable of improving workout accuracy, engagement, and performance tracking. Therefore, the following objectives are defined to address the identified problems:

- Build a portable sensor device using IMU technology that can accurately detect and count Reps as they happen during different strength exercises.
- Set up wireless communication through BLE so the sensor can send rep-counting information to a phone app quickly, with as little delay as possible.
- Develop a mobile app that works on both Android and iOS, where users can see their rep counts, set numbers, exercise information, timing details, and how well they're performing, all laid out in a way that's easy to understand and navigate.
- Create an interactive rhythm feature (like a mini game) that keeps users moving at the right speed during each exercise. This helps maintain proper TUT and prevents people from lifting too fast just to move heavier weights.

- Include detailed exercise tutorials with animated demonstrations showing users exactly how to set up and perform movements correctly. This reduces the chance of getting hurt and helps newcomers learn proper techniques faster.
- Automatically record workout history by saving all sets, reps, timing information, and exercise data without requiring users to manually type anything in. This makes it easier to analyze performance and track progress over time.
- Give immediate feedback during workouts about rep quality, whether the tempo is consistent, and if form is breaking down. Users can make corrections right away instead of finding out later.
- Generate summarized workout reports that allow users to monitor progress over daily, weekly, and monthly periods.

The system is aimed to be made reliable and flexible by being programmed to recognize different types of exercises, whether a barbell, dumbbells, or weight machines are used, through adjustable motion patterns. Personalized workout plans are allowed to be created based on what is to be achieved (building strength, increasing endurance, or growing muscle), with exercises and progressions that match those goals. Detailed set categorization is allowed, so each set can be labeled as a warm-up, a working set, a drop set, or a set taken to failure. More detailed information is therefore given when training patterns are analyzed.

1.4 Project Scope

To meet the goals that have been set while staying within reasonable development limits, a clear project scope has been defined that focuses on the core features needed to solve the main problems that were identified.

1.4.1. In Scope

The following section outlines the core functional deliverables and technical components that define the Slim as Fit. The project focuses on developing a fully integrated solution that combines embedded hardware, wireless communication, and mobile application development. The scope is specifically bounded to the features necessary to ensure accurate repetition counting, tempo analysis, and real-time user feedback for resistance training.

1. **Hardware Development:** A compact sensor device is developed using an accelerometer and a gyroscope to capture equipment movement during exercises.
2. **Wireless Communication:** The sensor is connected to a mobile application using BLE, allowing data to be transmitted in real time with minimal delay.
3. **Repetition Counting Algorithm:** Algorithms are developed to process motion signals and detect peaks to automatically count Reps as they occur.
4. **Tempo Analysis:** The duration of the lifting phase and lowering phase of each repetition is calculated and monitored by the system.
5. **Mobile Application:** A mobile application is developed for both Android and iOS platforms to track workouts, display performance data, and allow user interaction with the system.
6. **Rhythm-Based Training Feature:** An interactive, game-like feature is included to provide audio and visual cues that help maintain proper tempo.
7. **Exercise Tutorial System:** Animated demonstrations and clear, step-by-step instructions are provided to show exercises being performed correctly.
8. **Workout Logging:** Workout data including counting Reps, and time is recorded automatically. But the exercise with its weight values, no. of reps, sets and exercise plan (days/workouts) being the only data entered manually by the user.
9. **Performance Analytics:** Workout history is displayed through graphs and charts to support progress tracking over time.
10. **User Profile Management:** Personal accounts are supported, allowing fitness goals to be set and workout programs to be created based on desired outcomes.
11. **Exercise Type Support:** The system is designed to primarily support barbell lifts, dumbbell exercises, and standard weight machines.

1.4.2. Out of Scope

- **Advanced Biomechanical Analysis:** Detailed joint angles are not calculated, and complex movement models are not developed.
- **Vision-Based Form Correction:** Cameras are not used to analyze posture or body positioning.
- **Multi-Gym Deployment Infrastructure:** Commercial deployment systems are not built, and integration with gym management software is not included.

- **Nutrition Tracking:** Calories are not counted, meals are not planned, and dietary advice is not provided by the application.
- **Social Features:** Community forums, competitive leaderboards, and workout-sharing features are not included.
- **Wearable Heart Rate Integration:** Heart rate and cardiovascular data are not monitored during strength training workouts.

1.5 Limitations and Constraints

One limitation of the proposed system is the time required for users to become comfortable with its operation, as it introduces a new method for tracking workouts that may be unfamiliar, particularly within gym environments in Egypt. Additionally, the system depends on Bluetooth connectivity and a rechargeable battery, which introduces constraints such as potential signal interference, temporary disconnections, and battery depletion. These factors may occasionally interrupt data transmission between the sensor device and the mobile application, affecting continuous workout tracking.

1.6 Contributions / Significance of the Project

This project advances wearable fitness technology by improving how strength training is tracked. Instead of using traditional wrist-based wearables, a sensor is mounted directly on the exercise equipment, allowing more accurate tracking of Reps, movement patterns, and lifting tempo. This direct sensing approach leads to better performance measurement compared to current wearable devices.

A second contribution of this project is the use of real-time feedback to support safer and more effective training. Guidance is provided while the exercise is being performed, rather than only after it is finished. This helps users avoid poor lifting habits, encourages proper movement, reduces the risk of injury, and supports better muscle development.

In addition, the project demonstrates how embedded hardware, wireless communication, and mobile application development can be combined into one complete and integrated system. Automatic workout logging and clear presentation of

workout data help users stay consistent and motivated in their long-term training routines.

Finally, this project serves as a practical case study in designing and building a full smart system from start to finish. It shows how sensor data collection, real-time processing, and user-centered design can be applied together to solve a real-world fitness problem.

1.7 Project Organization

This documentation is organized as follows:

- **Chapter 2** investigates related works and existing solutions.
- **Chapter 3** presents system analysis and design.
- **Chapter 4** concludes with future work directions.

Chapter 2: Investigations

2.1. Overview

This chapter provides a market analysis of existing fitness applications and wearable solutions. It investigates current commercial products to identify their core functionalities, target demographics, and technical limitations. By examining these "related works," we establish the "data gap" in strength training technology, specifically regarding the lack of automated repetition counting, real-time tempo guidance, and form correction, which justifies the development of the Slim as Fit System.

2.2. Summary of related works

To understand the current landscape of fitness monitoring, we reviewed seven existing commercial solutions that are widely used by gym-goers today. We analyzed these systems based on their data acquisition method and whether they are an app or a website, and rely on manual user input, wrist-worn accelerometers, or dedicated hardware. This review identifies the specific technical shortcomings that the proposed Slim as Fit project aims to address.

2.2.1. JETFIT Gym Workout Tracker

JEFIT is a widely used mobile application designed for planning and logging gym sessions [12, 25].

- **Key Features:** It focuses on manual data entry, allowing users to log sets, reps, and weights for over 1,400 exercises. It features "One-Tap Workouts" to generate routines based on muscle groups.

- **Limitations:** The system relies entirely on user input. It does not utilize sensors to count Reps automatically or monitor lifting speed. It provides video instructions but offers no real-time feedback on the user's actual performance while they are lifting.

2.2.2. ZTFit Health

ZTFit Health is a companion application designed for budget smart bracelets [26].

- **Key Features:** Its primary function is general health monitoring, including step counting, sleep quality analysis, and heart rate monitoring. It supports multiple sport modes, but these are primarily cardio-focused (running, cycling) [19].

- **Limitations:** The application is limited to cardiovascular metrics and does not support strength training analytics. It cannot detect specific weightlifting movements or count Reps, and users frequently report connectivity and synchronization issues that would disrupt a real-time workout session [20].

2.2.3. Glory Fit Smart Wear

Glory Fit serves as a synchronization tool for generic smartwatches to log daily activity data [21, 27].

- **Key Features:** It tracks standard physiological metrics such as blood oxygen, heart rate, and sleep patterns. It offers post-workout visualization for calorie consumption and distance.

- **Limitations:** The system is "Cardio-Focused," optimized for steps and distance rather than stationary resistance training. It does not provide any "Rhythm Game" or tempo guidance features, making it incapable of correcting a user's lifting speed or form in real-time.

2.2.4. Hevy - Gym Log & Workout Tracker

Hevy is a popular manual logging application that emphasizes social interaction and routine planning [28].

- **Key Features:** It allows comprehensive workout logging (sets, reps, RPE) and features a "Custom Routine Builder" for advanced users. It includes a strong social component where users can share routines and view leaderboards.
- **Limitations:** Despite its popularity, Hevy operates without automatic detection; it cannot verify if a rep was actually performed or if the range of motion was complete, instead user has to do manual counting [18]. It offers no real-time

coaching or tempo analysis, relying solely on the user to judge their own performance.

2.2.5. Strong Workout Tracker

Strong is one of the most popular manual logging applications on the market, favored for its minimalist interface and speed [24, 29].

- **Key Features:** It focuses on efficiency, allowing users to build custom routines and log sets, reps, and Rate of Perceived Exertion (RPE) very quickly. It includes a "Plate Calculator" and charts for volume and One Repetition Maximum (1RM) progression over time.

- **Limitations:** Strong is strictly a passive data logger. It has no sensor integration and cannot verify if the user actually performed the exercise. It relies entirely on the user's honesty and memory, offering no real-time feedback on tempo or form consistency.

2.2.6. Fitbod (AI Planner)

Fitbod utilizes artificial intelligence (AI) to construct personalized workout plans based on the user's available equipment and past recovery data [30].

- **Key Features:** Its main strength is algorithmic planning; it tells the user exactly what to do (exercises, sets, reps) to achieve hypertrophy or strength goals. It adapts future workouts based on logged performance [10].

- **Limitations:** While it excels at planning, it fails at monitoring. The app assumes the user performs the recommended exercises correctly. It does not track the quality of the movement, the lifting speed (tempo), or the time under tension. The AI is limited to scheduling, not biomechanical analysis [10, 23].

2.2.7. Garmin Strength Features (Wearable)

Garmin (e.g., Fenix/Venu series) attempts to bridge this gap by using wrist-worn accelerometers to count reps automatically [31].

● **Key Features:** The "Strength" activity mode attempts to detect the type of exercise (e.g., bench press vs. curl) and counts Reps using wrist motion. It tracks heart rate and rest intervals between sets [22].

● **Limitations:** Wrist-based detection is often inaccurate for exercises where the wrist remains stable (e.g., leg press, squats). Users frequently report "missed reps" or "phantom reps," requiring them to manually edit the data after every set. Furthermore, it only counts Reps; it does not guide the user on lifting tempo or form quality.

2.3. Comparison between Related Works

When researching and comparing between existing products and technologies we managed to create a Comparison table as shown in Table 2.1 a comparison between the analyzed solutions against the core objectives of the proposed in Slim as Fit project.

Table 2.1 Comparison between related works

Feature	Strong App	Fitbod	Garmin Strength	JITFIT	ZTFit	Glory Fit	Hevy
Primary Focus	Manual Log	AI Planning	Cardio/GPS	Manual Log	Health	Health	Social
Auto-Rep Count	-	-	Yes (Wrist)	-	-	-	-
Tempo Analysis	-	-	-	-	-	-	-
Real-Time Feedback	-	-	-	-	-	-	-
User Input	Manual	Manual	Wrist Watch	Manual	Watch	Watch	Manual
Gamification	-	Badges	Badges	Social	-	-	Social

Our investigation of eight related works identifies critical limitations across three sets of categories. Manual apps (JITFIT, Hevy, Strong, Fitbod) serve as passive logs that fail to verify repetition counts or execution quality. Wearables (ZTFit, Glory Fit, Garmin) are optimized for cardio and suffer from accuracy issues in strength training due to wrist placement. Finally, most of them lack the gamification required to keep general users engaged.

This confirms a significant gap in the market: no existing solution combines automated tracking with pedagogical guidance. Slim as Fit addresses this by integrating precision equipment-mounted sensors with a gamified mobile interface, providing real-time feedback and safety checks currently missing from commercial products.

Chapter 3: System Analysis and Design

3.1. Overview

This chapter outlines the comprehensive analysis and design of the Slim as Fit app. Building upon the problem definition and objectives established in Chapter 1, this section details the system's architecture. It defines the specific functional and non-functional requirements necessary to bridge the gap in strength training monitoring. Furthermore, it presents the system design through Unified Modeling Language (UML) diagrams, including Use Case, Sequence, and Entity Relationship diagrams to visualize the interaction between the hardware sensor, the mobile application, and the backend infrastructure.

3.2. Methodology

The Slim as Fit utilizes a custom wearable sensor to capture motion data for rep-counting, tempo analysis, and real-time feedback via a mobile application.
By mounting the Inertial Measurement Unit (IMU) directly onto resistance equipment, the system bypasses the inaccuracies of wrist-based tracking, allowing for the precise correlation of raw kinematic signals with actionable, real-time coaching cues.

3.2.1. Hardware Architecture Design

The system uses a compact module attachable to the body or equipment, comprising:

- **Inertial Measurement Unit (IMU):** (Accelerometer and Gyroscope), It detects rep patterns, trajectory, tempo, and range of motion.
- **Microcontroller Unit (MCU):** Responsible for real-time signal processing, noise filtering, peak detection, and app communication.
- **Wireless Communication:** Bluetooth ensures fast, stable, and power-efficient data transmission to Android and iOS devices.
- **Power System:** A rechargeable Li-ion battery featuring gesture controls (tap/hold for operation), sleep mode, and motion-based auto-wake.
- **Form Factor:** A sweat-resistant, ergonomic, 3D-printed case housing component featuring a button for gesture controls, strapped using Velcro or magnets for stability during high-impact workouts.

3.2.2. Data Acquisition Pipeline

- **Motion Capture:** IMU collects high-frequency raw acceleration and angular velocity data.
- **On-Device Preprocessing:** MCU applies low-pass filtering and peak detection to segment cycles and identifies start/end points.
- **Transmission:** Processed metrics (rep count, orientation, tempo) are sent to the app via Bluetooth.

3.2.3. Calibration

The device ensures accuracy through zero-offset accelerometer calibration and exercise-specific motion data.

3.2.4. Hardware and Software Integration

The system employs hybrid processing:
Device: Handles lightweight calculations (feature extraction).
App: Performs advanced classification and user feedback. It follows the flow shown in Figure 3.1.

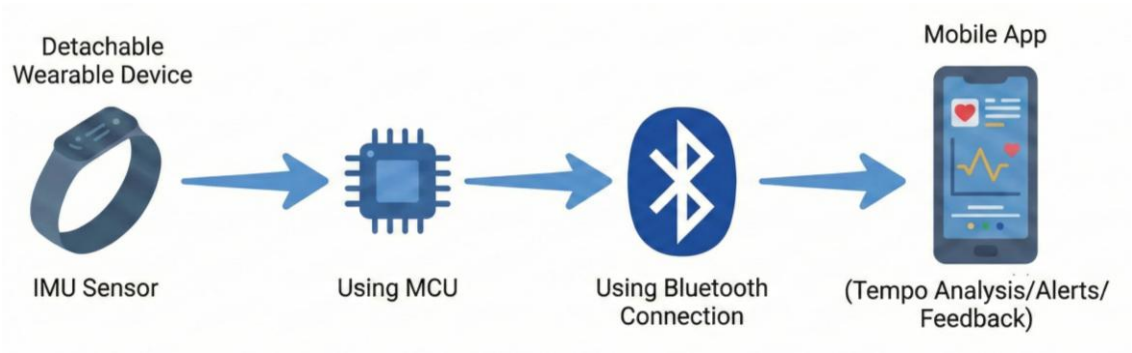

Figure 3.1 Hardware and Software Integration Diagram

3.2.5. Prototype Development

Electronics: Component selection (IMU/MCU), circuit design, and Printed Circuit Board (PCB) / breadboard assembly.

- **Firmware:** Development of IMU drivers, motion algorithms, BLE protocols, and power management.
- **Enclosure/housing:** 3D printed design focusing on heat ventilation and velcro strap connectors.

- **Testing:** Validated through accuracy trials, dynamic workout stress tests, and latency evaluation.

3.2.6. Key Innovations

- **Specificity:** Focuses on correct form/tempo per exercise rather than general **activity** tracking.
- **Precision:** Optimal sensor placement ensures higher accuracy than standard **smartwatches**.
- **Real-Time Feedback:** Provides immediate correction rather than **passive** data logging.
- **Gamification:** Enables the Rhythm Rep Game by converting sensor data into **gameplay** mechanics.

3.2.7. Scientific backing

The core functionality of the system, specifically its emphasis on rep-counting and real-time tempo analysis, is heavily grounded in the biomechanical principles of muscle hypertrophy. To achieve optimal muscle growth, it is not simply the amount of weight lifted that matters, but *how* the weight is moved.

- **Time Under Tension (TUT) and Muscle Activation:** Hypertrophy requires muscles to be placed under continuous mechanical stress for a specific duration [4]. By lifting a weight slowly and releasing it slowly with controlled form, the lifter eliminates the use of momentum. This deliberate pacing forces the entire targeted muscle group to activate, maximizing the recruitment of muscle fibers rather than relying on joints or secondary muscles to "cheat" the movement [6].
- **Importance of Concentric and Eccentric Phases:** A complete repetition consists of two main phases: the concentric phase (lifting the weight and contracting the muscle) and the eccentric phase (releasing or lowering the weight and lengthening the muscle). Scientific literature indicates that the eccentric phase is just as critical if not more so for inducing the micro-tears in muscle tissue required for growth [5]. The system monitors these phases independently to ensure the user does not neglect the lowering portion of the lift.
- **Optimal Tempo Windows:** For effective hypertrophy, a single repetition must fall within a specific time threshold (typically a minimum of 2 seconds and a

maximum of 6 to 8 seconds per rep, combining both lifting and lowering) [10]. The system uses the IMU sensor to track this exact duration:

Moving Too Fast: If a repetition is performed below the minimum time threshold (moving too fast), the user relies on kinetic momentum rather than muscle strength. This drastically reduces the physiological benefits of the exercise and significantly increases the risk of tendon and joint injuries due to jerky, uncontrolled motions.

Moving Too Slow: Conversely, if the repetition exceeds the maximum time threshold, it indicates an inability to sustain the load. Prolonged, excessively slow repetitions lead to rapid neuromuscular fatigue. When the tempo drops below this critical threshold, it signals that the user has reached temporary **muscle failure** and should safely terminate the set to prevent overtraining or failure-induced injury [3,10].

3.3. Functional and Non-functional Requirements

This section defines the operational specifications and performance constraints necessary for the successful deployment of the Slim as Fit system. The requirements are categorized into Functional Requirements (FRs), which describe the specific behaviors and capabilities the system must execute, and Non-functional Requirements (NFRs), which establish the quality attributes such as latency, accuracy, and power efficiency that determine how well the system performs its functions.

3.3.1. Functional Requirements

Functional Requirements (FRs) These requirements define the specific behaviors and functions the system must support, based on the project objectives and Use Case definitions:

- **FR-01 Repetition Counting:** The system must utilize IMU data to detect and count Reps automatically as they occur.
- **FR-02 Tempo Monitoring:** The system must calculate the duration of the concentric (lifting) and eccentric (lowering) phases to monitor TUT.

- **FR-03 Real-Time Feedback:** The mobile app must provide immediate visual or audio cues (e.g., "Rhythm Rep Game") to guide the user's lifting speed and form.
- **FR-04 Workout Logging:** The system must automatically record sets, reps, and timing data without manual input, saving it to the database.
- **FR-05 User Authentication:** Users must be able to register, login, and manage their profiles securely.
- **FR-06 Exercise Tutorials:** The app must display animated demonstrations and instructions for proper exercise execution.

3.3.2. Non-functional Requirements

These requirements specify the quality attributes and performance standards:

- **NFR-01 Latency:** Data transmission between the sensor and the app must occur with minimal delay to support real-time rhythm cues.
- **NFR-02 Accuracy:** The rep-counting algorithm must accurately differentiate between actual Reps and random equipment movements.
- **NFR-03 Battery Efficiency:** The sensor module must utilize power-saving features like sleep mode and motion-based auto-wake to prolong usage.
- **NFR-04 Ergonomics**: The sensor enclosure must be compact, sweat-resistant, and easily attachable to various equipment types (barbells, dumbbells) using Velcro or magnets.
- **NFR-05 Connectivity:** The system must maintain a stable Bluetooth connection in a gym environment, potentially prone to signal interference.

3.4. Software and Hardware requirements

This section details the technical specifications and operational resources necessary for the successful implementation of the Slim as Fit project. The requirements are categorized into **Hardware**, which lists the physical components required to construct the sensor node, and **Software**, which outlines the development frameworks, databases, and platforms used to build the mobile application and backend infrastructure.

3.4.1. Hardware Requirements

The physical implementation of the sensor node relies on the following components:

- Inertial Measurement Unit: Combined Accelerometer and Gyroscope for detecting motion trajectories.
- Microcontroller Unit: For signal processing, noise filtering, and managing the BLE stack.
- Communication Module: BLE module for wireless data transfer.
- Power Supply: Rechargeable Li-ion battery.
- Enclosure: 3D-printed case designed for heat ventilation and strap attachments (e.g., Velcro straps).

3.4.2. Software Requirements

- Mobile Application: Developed for Android and iOS (e.g., using Flutter) to serve as the user interface.
- Database: A structured database Structured Query Language (SQL) to store user profiles, workout plans, and sensor logs.
- Firmware: Embedded software for the MCU to handle IMU drivers, motion algorithms, and power management.

3.5. System Design

This section outlines the integration of the sensor and mobile app with the backend. The following UML diagrams serve as the primary visual definitions for the system's data flow and operation sequence and real-time monitoring logic.

3.5.1. System Architecture

Figure 3.2 illustrates the high-level architecture of the Slim as Fit solution, depicting the end-to-end data flow from physical motion to digital analytics. The system operates across three primary layers: the Perception Layer, where the equipment-mounted sensor utilizes an IMU and MCU to capture and pre-process raw motion signals; the Communication Layer, which employs BLE for efficient, real-time transmission; and the Application Layer, where the mobile app executes the "Digital Spotter" logic to provide immediate feedback while synchronizing long-term performance data with the backend database.

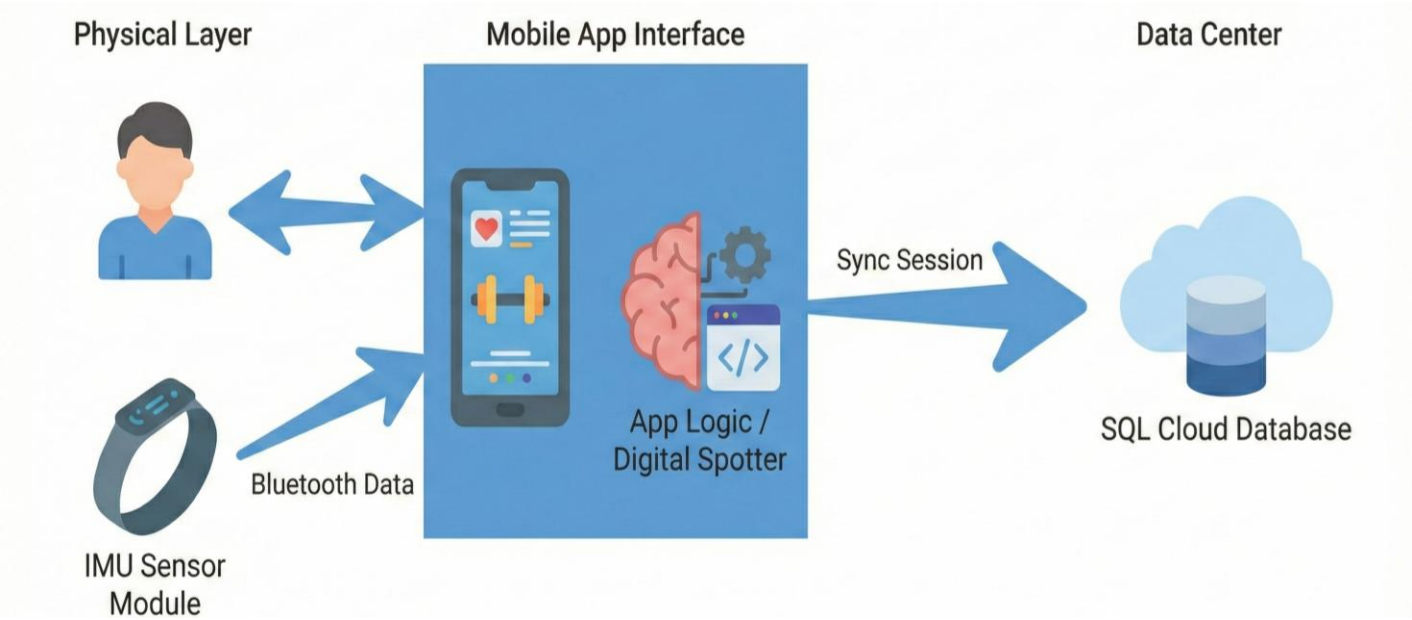

Figure 3.2 System Architecture Diagram

3.5.2. Use Case diagram

The system's functional scope is visualized through the Use Case Diagram shown in Figure 3.3.

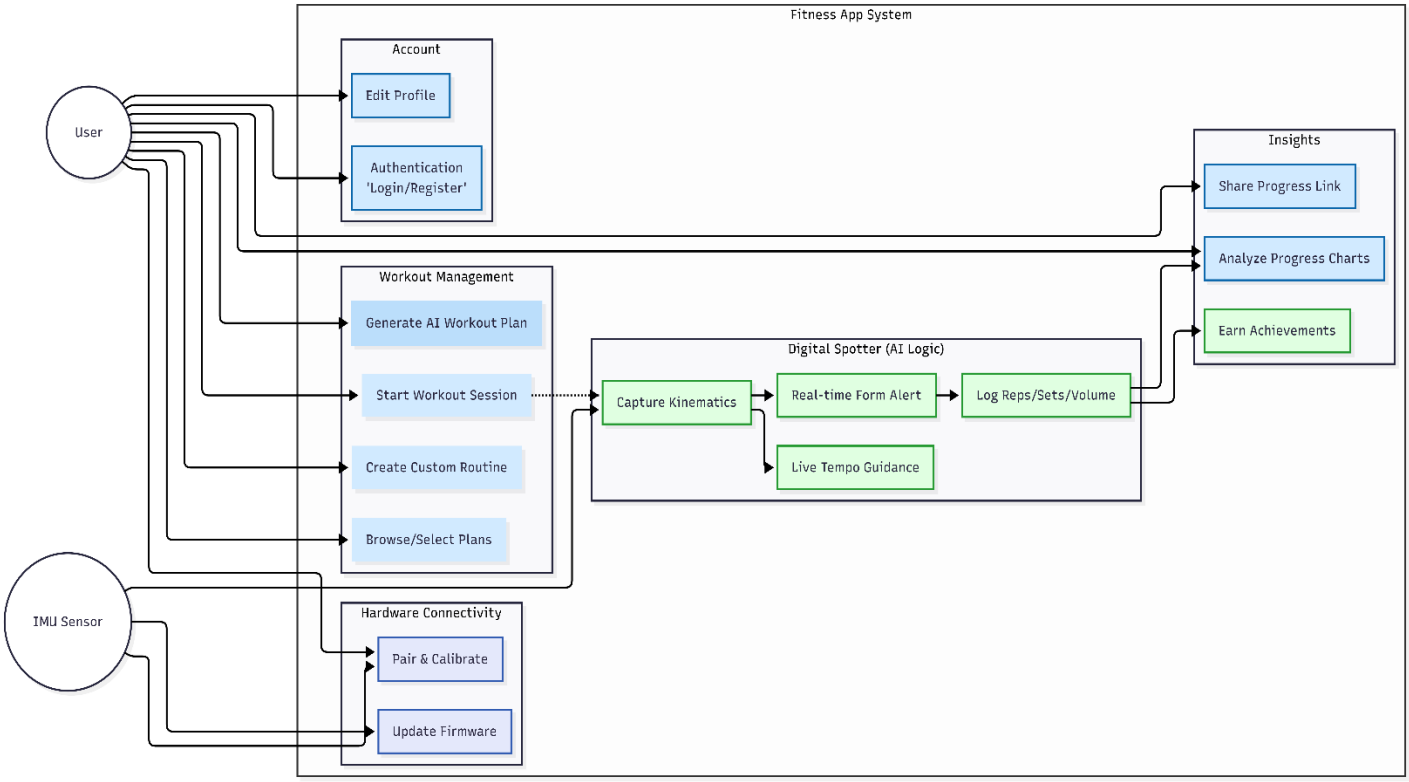

Figure 3.3 Use Case Diagram

The use case diagram identifies the "User" as the primary actor interacting with the "Fitness App System" and the "IMU Sensor." Key use cases include:

- **Account Management:** Authentication (Login/Register) and Profile Editing.
- **Workout Management:** Creating custom routines, browsing plans, and starting sessions.
- **Digital Spotter (AI Logic):** The core intelligence that captures **kinematics**, provides live tempo guidance, and issues real-time alerts.
- **Hardware Connectivity:** Pairing, calibrating, and updating firmware for the IMU sensor.
- **Insights:** Analyzing progress charts and earning achievements.

3.5.3. Sequence Diagram

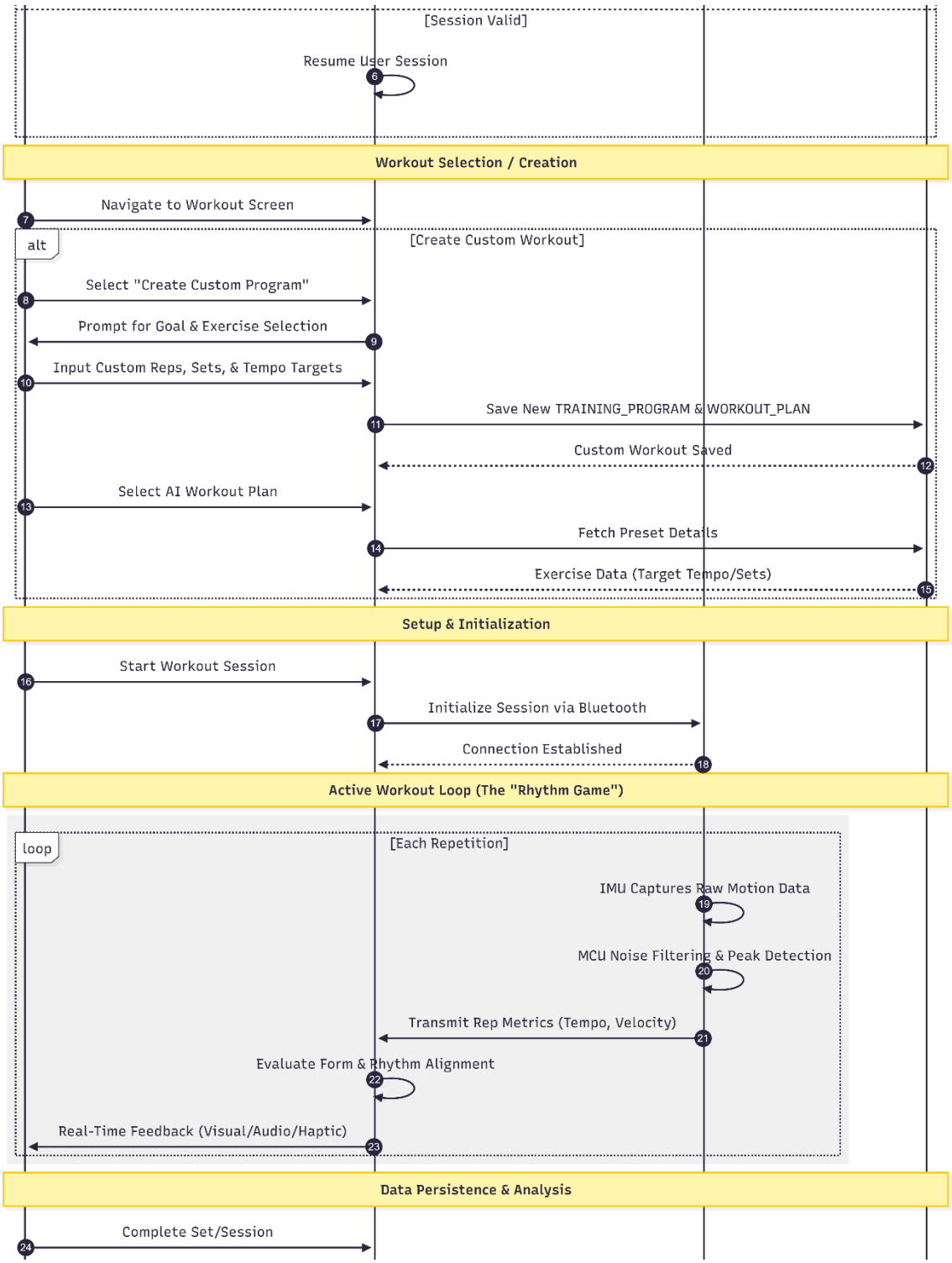

Figure 3.4 Sequence diagram

Illustrated in Figure 3.4 is the Sequence Diagram that models the logical flow of operations during a workout session. The process begins with Authentication, where the application validates user credentials against the database. This is followed by Workout Selection, in which the user selects a predefined workout program or creates a custom one. The system then enters the Active Workout Loop, during which the user performs Reps while the IMU sensor captures raw motion data. The MCU processes this data by filtering noise and detecting peaks, then transmits key metrics such as tempo and velocity to the mobile application. The application evaluates rhythm alignment and provides real-time visual and audio feedback to the user. Finally, during Data Persistence, once a set is completed, the application stops data acquisition and stores the WORKOUT_SESSION and SENSOR_DATA in the database.

3.5.4. Activity Diagram

The Activity Diagram illustrated in Figure 3.5 delineates the operational workflow of the Slim as Fit application, structuring the user journey from initial authentication to performance analysis. The process begins with session validation, leading to the dashboard where users select or configure specific routines and workouts. The core of the system is the "Active Workout Loop," a continuous cycle where the sensor captures raw motion data while parallel processes analyze tempo and range of motion to generate immediate audio-visual feedback. Upon completing a set, the system exits this loop to execute data persistence protocols, saving high-fidelity sensor logs to the database before finally presenting the user with synthesized progress.

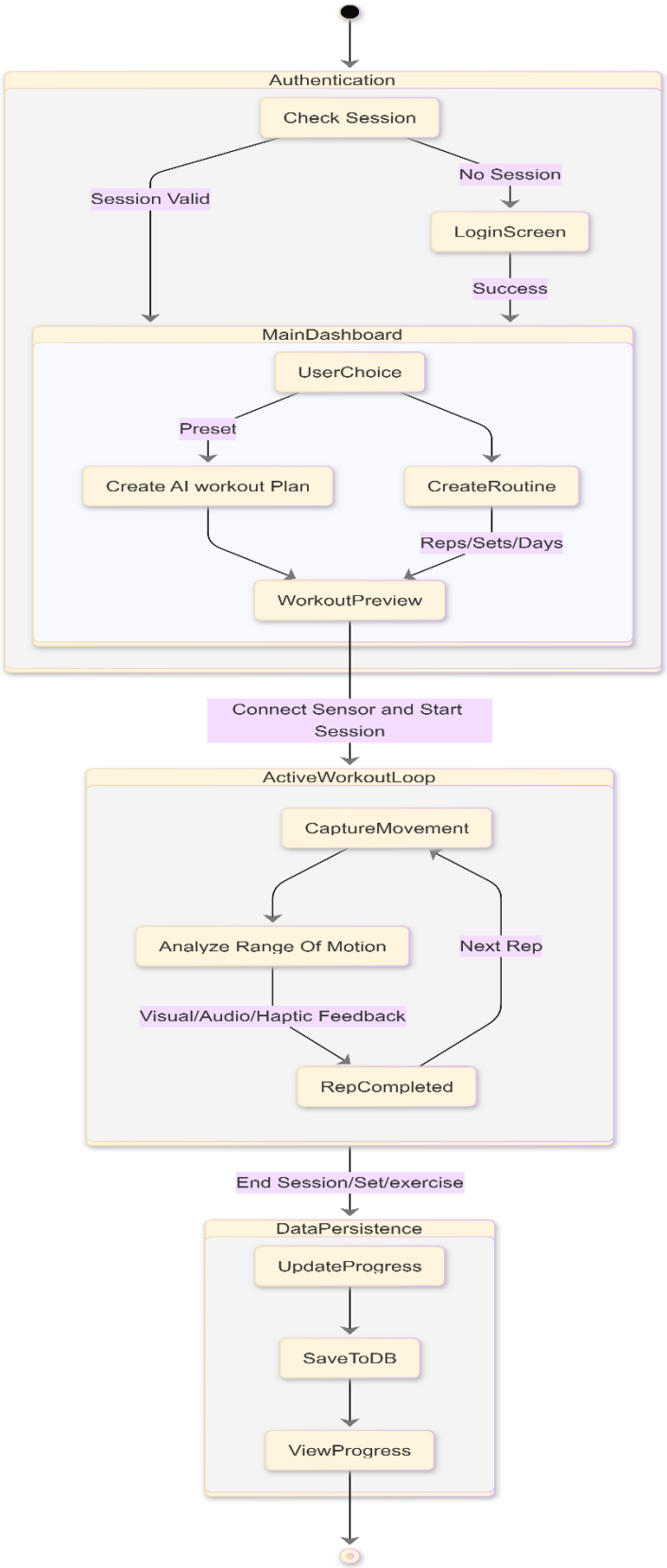

Figure 3.5 Activity Diagram

3.5.5. Entity Relationship Diagram

Figure 3.6 is the Entity Relationship Diagram (ERD) defines the relational database structure designed to manage user progression and high-frequency sensor telemetry within the Slim as Fit ecosystem. The schema centers on the User entity, which maintains relationships with **the Training Program** and **Workout Session** tables to organize fitness routines and historical logs. Crucially, the **Exercise Set** entity acts as the fundamental unit of data recording, linking specific physical activities directly to the **Sensor Data** table which stores raw IMU metrics and the **Performance Feedback** table, ensuring that every repetition is accurately indexed against its corresponding biomechanical analysis.

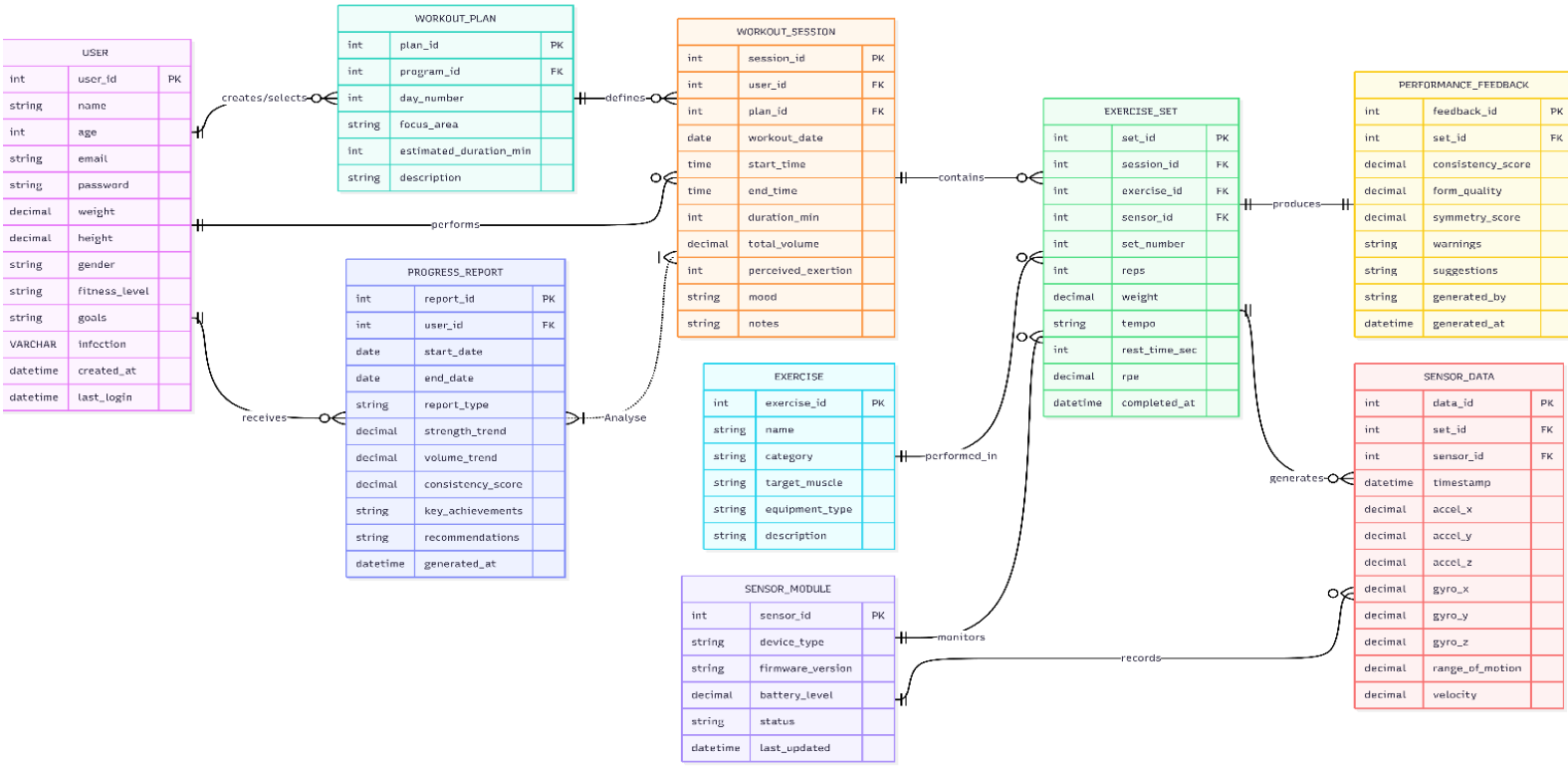

Figure 3.6 Entity Relationship Diagram (ERD)

3.5.6. User Interface Deign

The user interface is designed to be intuitive and minimize distraction during exercise, with an energetic blue and white color palette.

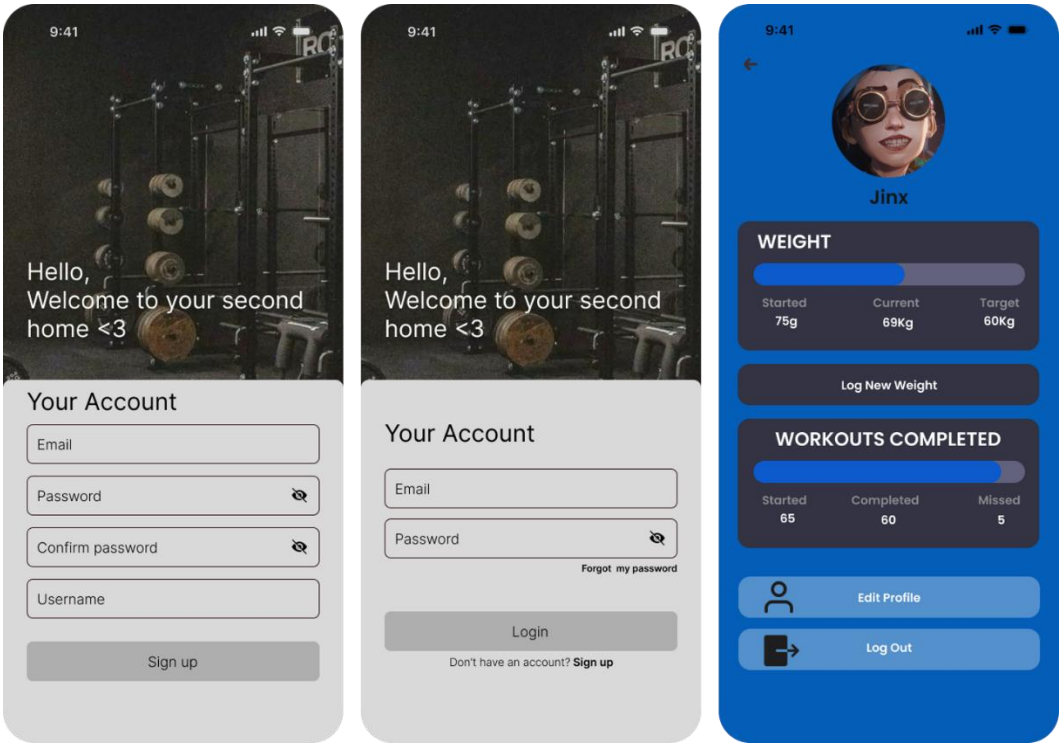

Figure 3.7 User Interface Login Form Page & Profile

These screens (Figure 3.7) show the main flow of the fitness application. The first screen allows new users to sign up by creating an account, while the second screen is designed for existing users to log in quickly and easily. After logging in, the third screen presents the user portfolio, where users can view their profile information, track their weight progress, and monitor completed workouts. The overall design is clean and welcoming, helping users feel comfortable and motivated while using the app.

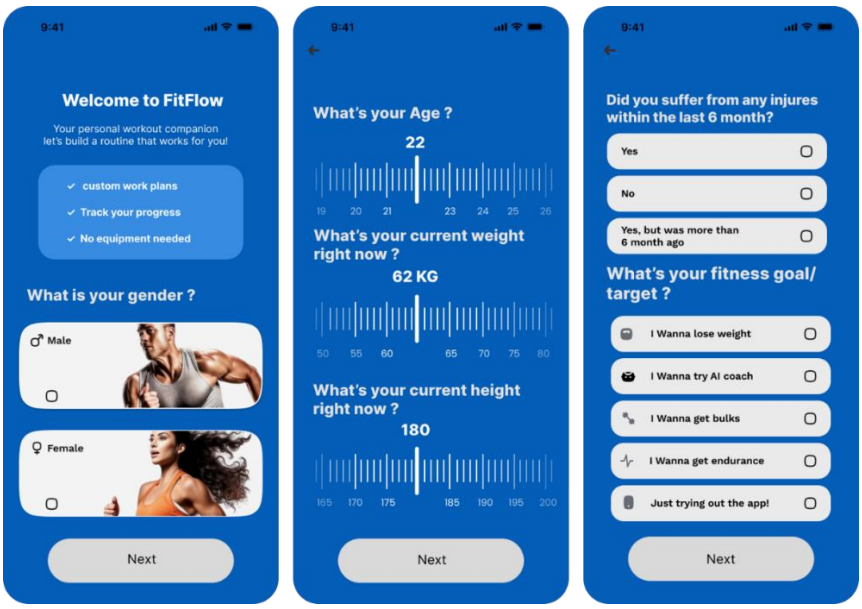

Figure 3.8 User Initial Setup Form Page

These screens (Figure 3.8) guide the user through the initial onboarding process in a simple and friendly way. The first screen introduces the app and its features, while allowing the user to select their gender. The second screen collects basic personal information, including age, current weight, and height, using easy-to-adjust sliders. The third screen focuses on the user's fitness background and preferences by asking about any recent injuries and letting the user choose their fitness goal from clearly labeled icons. This step-by-step flow helps personalize the workout experience from the start.

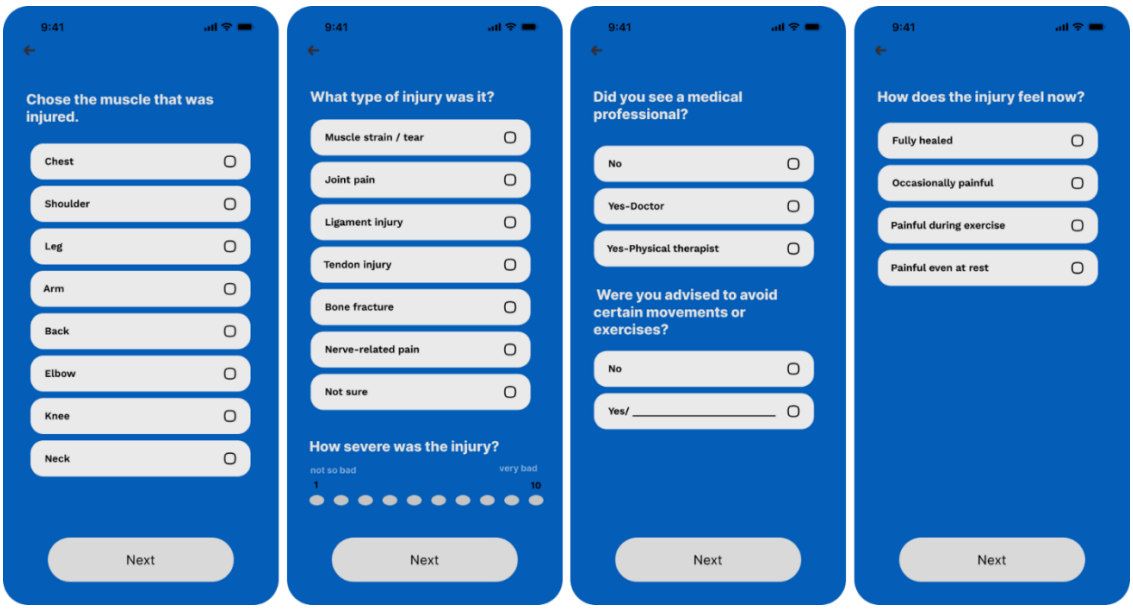

Figure 3.9 User Injury Form

These screens (Figure 3.9) show a step-by-step injury intake form designed to collect clear and relevant information from the user. The flow guides the user through selecting the injured body area, identifying the type and severity of the injury, and indicating whether a medical professional was consulted. It also captures any movement restrictions and the user's current pain level. Overall, the layout keeps each step simple and focused, making it easy for users to report their injury accurately without feeling overwhelmed.

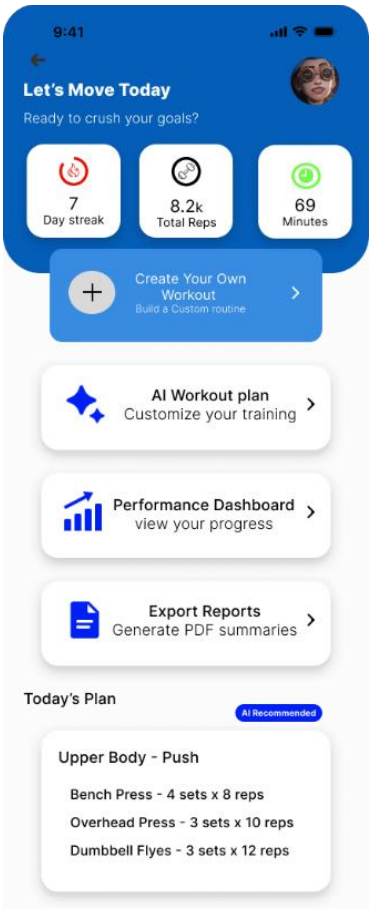

Figure 3.10 Home Screen

This screen (Figure 3.10) presents the main dashboard of the fitness app, giving users a quick snapshot of their progress and daily activity. It highlights key metrics such as workout streak, total reps, and time spent training, helping users stay motivated at a glance. From here, users can easily create a custom workout, follow an AI-generated training plan, track performance through detailed dashboards, or export reports. The layout is designed to feel encouraging and intuitive, guiding users smoothly into their workout for the day.

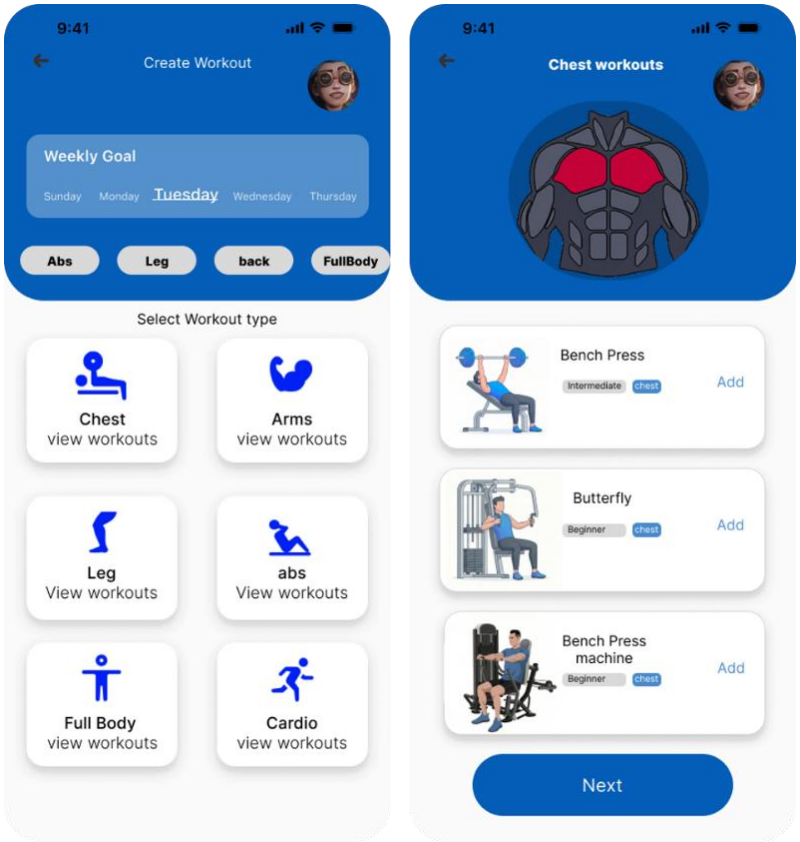

Figure 3.11 Create Workout Screen

These screens (Figure 3.11) illustrate the workout creation flow, where users can plan their training based on weekly goals and targeted muscle groups. The first screen allows users to choose a workout category such as chest, arms, legs, or full body, making it easy to tailor sessions to their preferences. Once a muscle group is selected, the next screen displays relevant exercises with difficulty levels and quick add options. This guided setup helps users build effective workouts while keeping the experience simple and personalized.

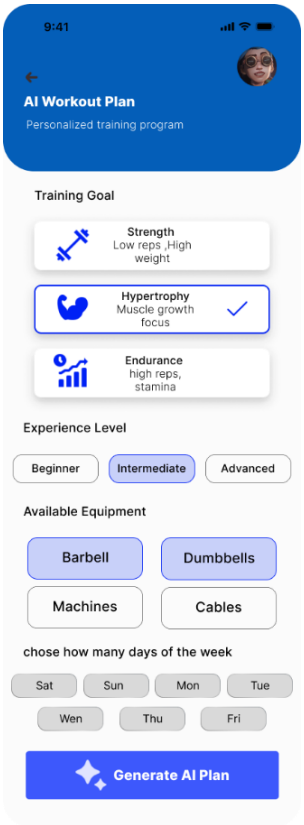

Figure 3.12 AI Workout Plan Screen

This Screen (*Figure 3.12*) shows an option to create workout plan based on your need; Goal, Skill level, Equipment variable and no. of days.

Figure 3.13 Progress Tracking Screen

This screen (Figure 3.13) shows progress throughout the weekdays, where the user can see deep analytics for each day.

Figure 3.14 Progress Analytics Screen

This screen (Figure 3.14) shows and example of how analytics could look like; showing tempo analysis no. Of reps done and workout duration.

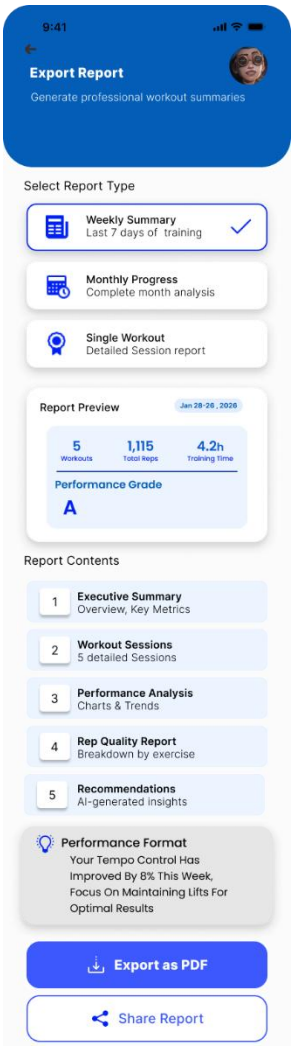

Figure 3.15 Progress Export Screen

This screen (Figure 3.15) shows an example of exporting the user's data so that it can be sent to a trainer, showing weekly or monthly or even a single workout session, then export as pdf or share link.

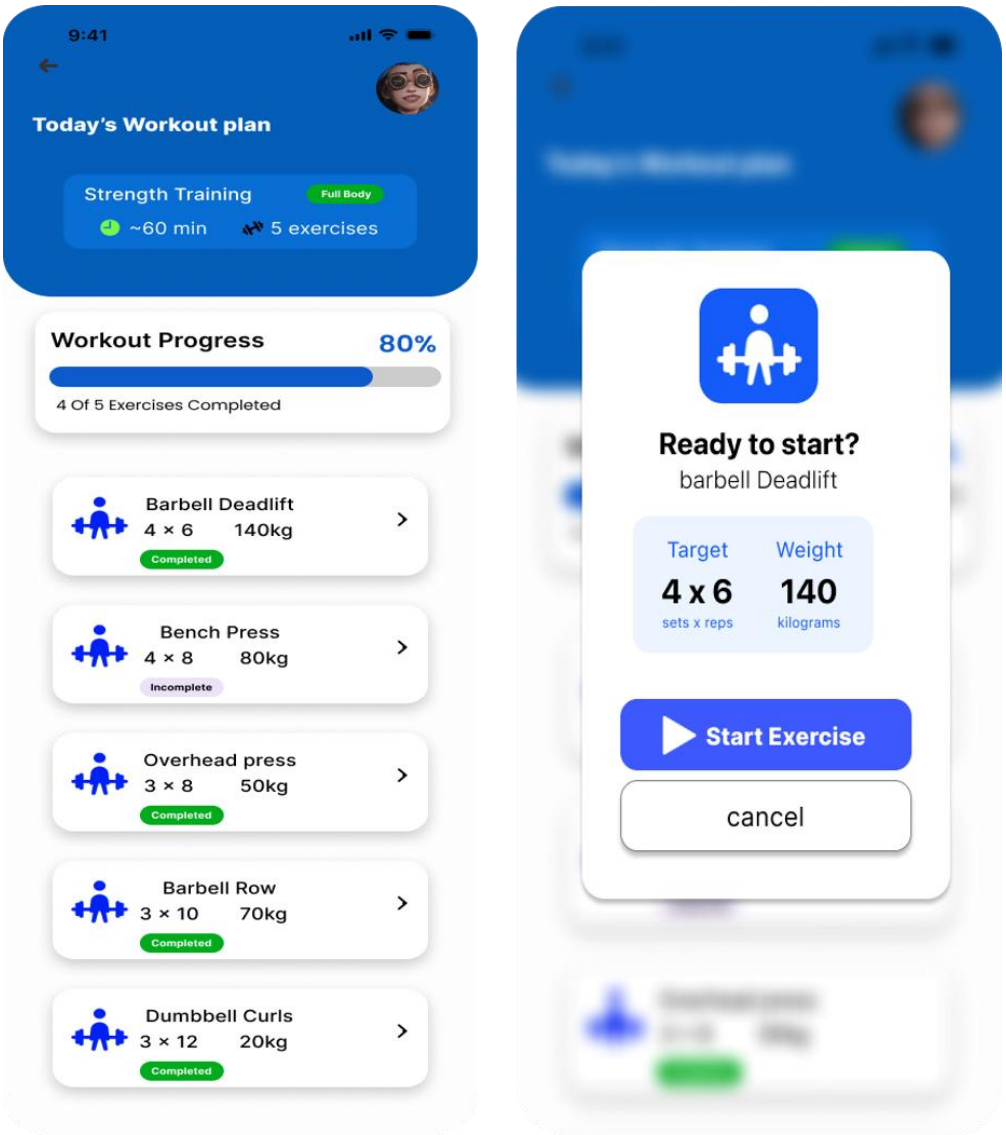

Figure 3.16 Workout Screen Overview

This screen (Figure 3.16) shows how the workout would look like showing every exercise and its weight, no. of sets and reps.

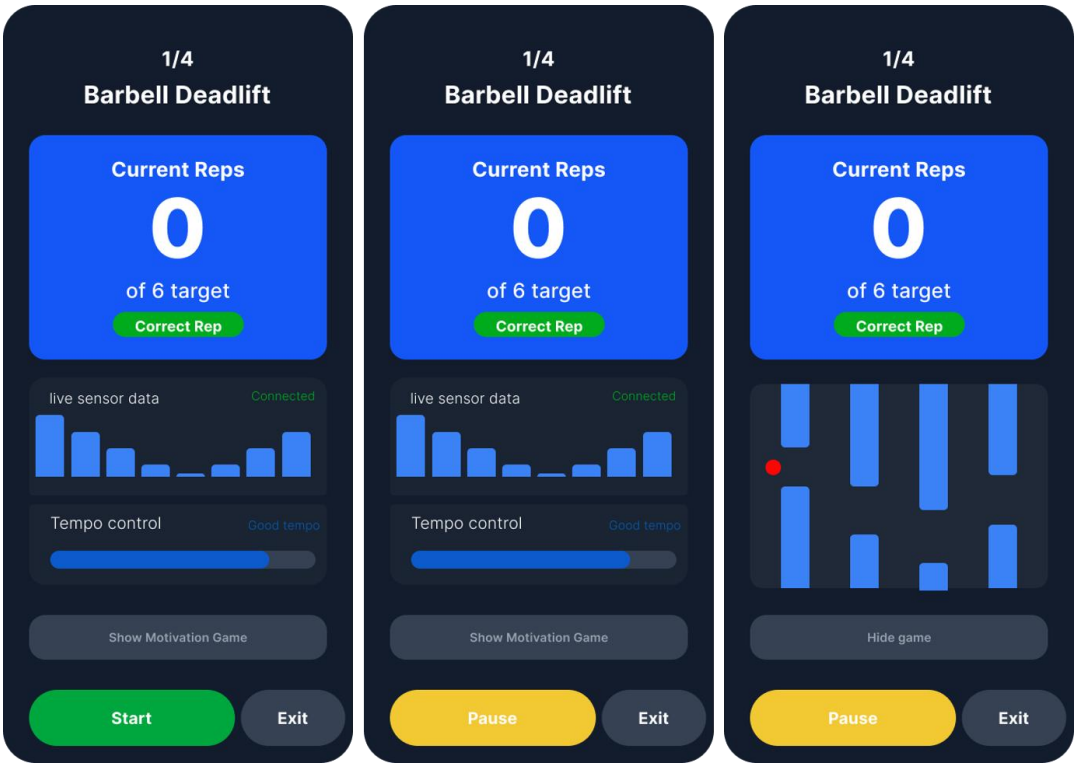

Figure 3.17 Active Workout Screen

This screen (Figure 3.17) shows a gamification example showing a flappy bird like game where the user must follow the correct tempo or lose; encouraging a correct workout tempo that's not too fast nor too slow on acentric and eccentric movements.

Figure 3.18 Workout Illustration Examples

This (Figure 3.18) is an example of the workout illustration showing starting position and middle pose so that they can alternate to make a makeshift video that shows how the exercise is done, so that when the user sees an exercise that uses the equipment the user has in his gym or exercise space the user picks this workout and add to your plan.

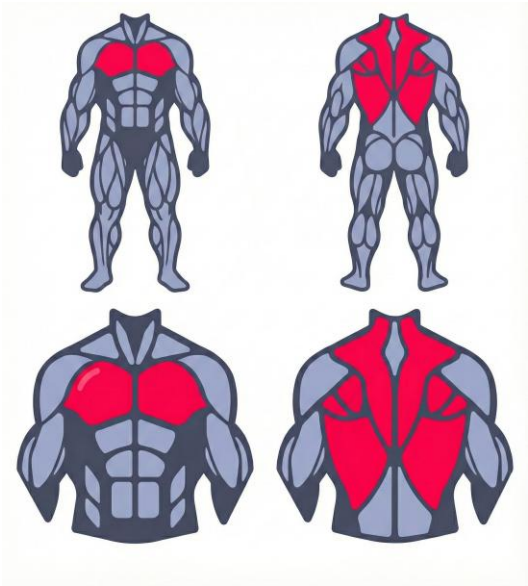

Figure 3.19 Muscle Select UI Element Example

This (Figure 3.19) is a UI element example of muscle selection, where the user can select the muscle or muscle group he wants to train in his workout plan day.

3.6. Feasibility Study

This section evaluates the practical feasibility of the proposed Slim as Fit solution to ensure it can be successfully implemented within the project's constraints. The study assesses the system's potential from three critical perspectives: technical feasibility, regarding the availability of required hardware and software resources; operational feasibility, concerning user adoption and integration into daily gym routines; and economic feasibility, analyzing the cost-effectiveness of the device compared to current market alternatives.

3.6.1. Technical Feasibility

The project is technically feasible as it utilizes established technologies. The availability of compact IMUs and low-power MCUs allows for the creation of the

sensor device. The algorithms required for peak detection and signal processing are well documented in wearable technology literature.

3.6.2. Operational Feasibility

The system addresses a validated need for strength training monitoring. By automating data logging, the system reduces the manual burden on the user, increasing the likelihood of adoption. The device is designed to be portable and easily mountable, fitting seamlessly into existing gym routines without requiring complex infrastructure.

Compared to the recurring cost of a personal trainer, Slim as Fit offers a cost-effective "digital spotter" solution. The use of standard electronic components and 3D printing for prototyping keeps development costs manageable while providing high value to the user through advanced metrics and safety features.

Chapter 4: Conclusion and Future Work

4.1. Conclusion

The proposed system successfully bridges the data gap in strength training by moving away from the current approach of using wrist-worn devices to a more direct approach of equipment-mounted sensors. By focusing on Time Under Tension (TUT) and precise Rep Counting, SLIM as FIT offers a specialized solution for muscle hypertrophy that general fitness wearables currently do not provide. This project proves that capturing the motion of the weights provides much more meaningful insights for strength athletes than standard activity trackers.

4.2. Limitations

Even though the proposed system has promising features and capabilities, it still has some limitations and dependencies that might affect its performance, including:

- **Cardiovascular Data:** The present version of the system only focuses on the kinematics of movement, or reps and tempo, and doesn't track any information on the physiological state of the user, such as heart rate or calories burned.
- **Bluetooth and Distance Reliability:** The system depends on a reliable Bluetooth connection between the equipment sensor and the phone. However, Bluetooth can be interfered with in a busy gym environment.
- **Manual Entry:** While the system can automatically track reps for exercises, users still need to manually enter their weight values.

4.3. Future work

To further improve the system's capabilities without compromising the simplicity of the system, the following enhancements can be made to the system in the future:

- **Smartwatch Integration:** integrating smartwatches with the Slim as Fit app. This will enable the system to read the heart rate and oxygen level data in real time during the set.
- **Intensity Analysis:** Utilizing heart rate data from the watch and the sensor's motion data to provide an accurate Rate of Perceived Exertion to the users to determine whether they are working efficiently.

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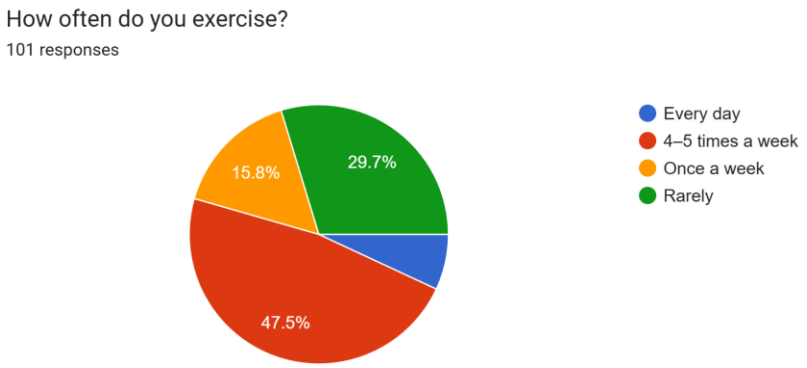
Appendix

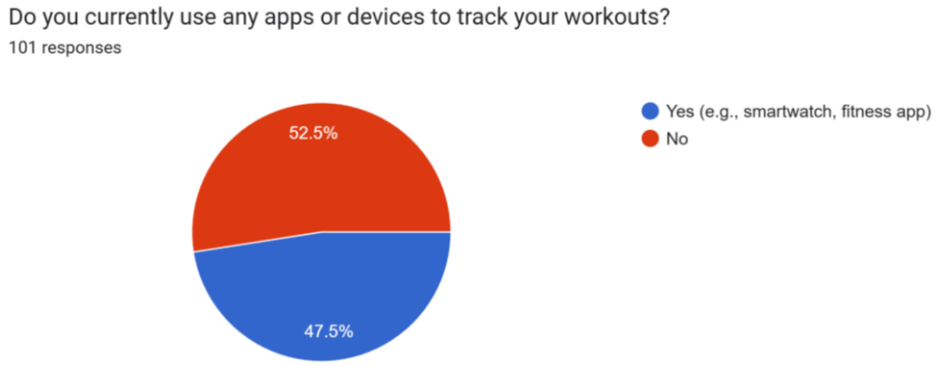
Appendix A.

Extra Survey Questions:

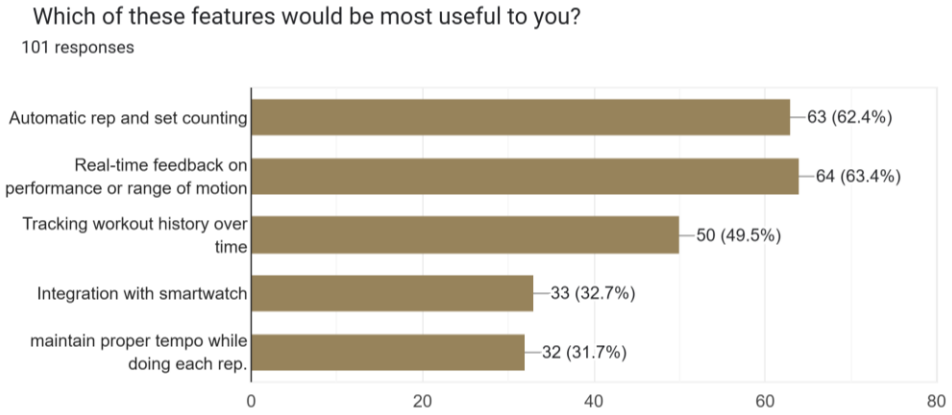
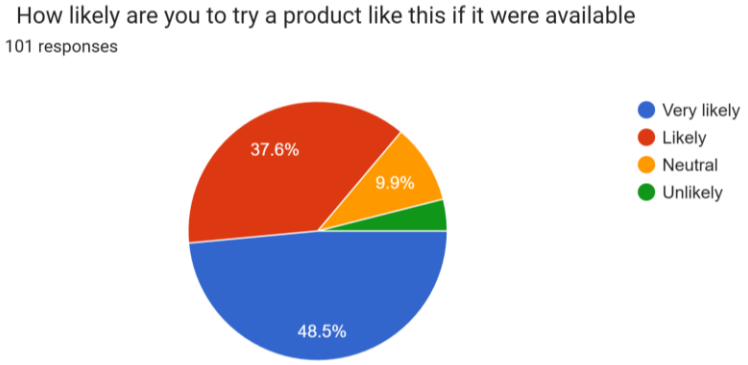
1. How often do you exercise?



2. Do you currently use any apps or devices to track your workouts?



3. Which of these features would be most useful to you?



4. How likely are you to try a product like this if it were available?